

SUPPLEMENTARY MATERIAL

A Multi-Source Atmospheric Refraction Framework for Geodetic Applications

Alkalali, K.Z.A. (2026) — Geodesy and Geodynamics (submitted)

This supplementary document provides full details of data sources, API parameters, file inventories, and complete datasets for all validation experiments reported in the main paper. All values are real computed or measured quantities — no values are assumed, estimated, or fabricated.

S1. Data Sources, API Parameters, and File Inventory

S1.1 ERA5 L137-Level Reanalysis

Parameter	Value
Dataset	reanalysis-era5-complete
Class / Type / Stream	ea / an / oper
Model levels	75 to 137 (63 levels, surface to ~20 km)
Variables	130 (temperature T, K), 133 (specific humidity q, kg/kg), 134 (surface pressure, Pa)
Horizontal resolution	0.25 x 0.25 degrees (~28 km)
Temporal resolution	6-hourly (00:00, 06:00, 12:00, 18:00 UTC)
S3 path (model levels)	kappa-era5-data/elite/raw/{YYYY-MM-DD}/{HHMM}_ml.nc (ERA5 L137 L137-level data)
S3 path (surface pressure)	kappa-era5-data/elite/raw/{YYYY-MM-DD}/{HHMM}_sp.nc (ERA5 surface pressure)
Regional extraction	6 regions: Middle East, Europe, Africa, Asia, Americas, Oceania
Regional boundaries	Africa: lat -40 to 40, lon -20 to 55; Americas: lat -60 to 80, lon -180 to -30; Asia: lat 0 to 75, lon 50 to 150; Europe: lat 20 to 75, lon -30 to 50; Mideast: lat 10 to 50, lon 25 to 75; Oceania: lat -50 to 10, lon 100 to 180
Global file size	~192 MB per time slot
Regional file size	7-66 MB depending on region
Citation	Hersbach et al. (2020)

Table S1. ERA5 L137 Copernicus CDS API request parameters.

S1.2 Complete S3 Data Inventory

The following atmospheric data sources are stored in Amazon S3 (bucket: kappa-era5-data, region: eu-north-1), updated daily via automated EC2 processes. Total storage: 1.0 TiB across 357,750 files as of June 2026.

Data Source	S3 Path Prefix	Date Coverage	Description
ERA5 L137 (137 levels)	elite/raw/{date}/	2012-06-01 to present (daily)	137 model levels; primary source for LOS and survey modes
ERA5 pressure levels	pro/raw/{date}/	2024-12-01 to present (daily)	37 pressure levels; alternative to L137 for survey applications
ERA5-Land	era5/land/{date}/	2012-06-01 to present (daily)	0.1 deg; skin temperature, 2m T, dewpoint, surface pressure
ERA5-Single level	era5/single/{date}/	2024-01-01 to present (daily)	Boundary layer height, TCWV, SST for specialist modes

GFS NWP	gfs/{date}/	Rolling 10-day window	0.25 deg forecast; Drone/UAV mode only
NOAA IGRA2 radiosonde	observations/radiosonde/{date}/	2025-01-01 to present (daily)	889 global stations; supplementary context

Table S2. Complete S3 data inventory for KAPPA Arc V3. Historical Saudi Arabia campaign dates (2012-06-01 to 2012-08-31), Spain campaign dates (2023-07-20, 2024-06-25), and tropical validation dates (2025-07-15 and 2025-01-15 at Barbados, Pohnpei, San Juan, and Singapore) are also stored for validation purposes. ICAO standard atmosphere is computed analytically — no storage required.

S1.3 ERA5-Land

Parameter	Value
Dataset	reanalysis-era5-land
Variables	t2m (2m temperature), d2m (2m dewpoint), skt (skin temperature), sp (surface pressure)
Horizontal resolution	0.1 x 0.1 degrees (~11 km)
Temporal resolution	6-hourly
S3 path	kappa-era5-data/era5/land/{YYYY-MM-DD}/{HHMM}.nc (ZIP format)
Surface anchor	skt at 0 m; t2m at 2 m above ground
Citation	Muñoz-Sabater et al. (2021)

Table S2. ERA5-Land parameters.

S1.4 GFS Numerical Weather Prediction

Parameter	Value
Source	NOAA Global Forecast System (GFS)
Resolution	0.25 x 0.25 degrees
Update cycle	4x daily
Application	Drone/UAV mode only
Access	NOAA NOMADS (nomads.ncep.noaa.gov)

Table S4. GFS data source parameters.

S1.5 NOAA IGRA2 Radiosonde Archive

Parameter	Value
Source	NOAA Integrated Global Radiosonde Archive, Version 2.2 (NOAA NCEI, 2023)

Active stations	889 globally, January 2025 onward
S3 path	kappa-era5-data/observations/radiosonde/{YYYY-MM-DD}/{StationID}_{HHMM}.json
JSON fields	station_id, station_name, lat, lon, elev, date, hour, levels, total_levels, source
Level fields	p_hpa, alt_m, temp_c, dewp_c
Surface level	Level with highest p_hpa value in sounding
Temperature	temp_c field — degrees Celsius directly
Update schedule	Daily: 03:00 UTC regional, 04:00 UTC global (889 stations)

Table S5. NOAA IGRA2 radiosonde archive structure.

S1.6 Saudi Arabia Validation — S3 File Inventory

Date	L137 Files	ERA5-Land Files	Status
1 June 2012	32	4	Verified
15 June 2012	32	4	Verified
30 June 2012	32	4	Verified
1 July 2012	32	4	Verified
15 July 2012	32	4	Verified
31 July 2012	32	4	Verified
1 August 2012	32	4	Verified
15 August 2012	32	4	Verified
31 August 2012	32	4	Verified

Table S6. ERA5 S3 file inventory for nine-date Saudi Arabia campaign. 32 L137 files = 8 global (4 time slots x 2 variables) + 24 regional (6 regions x 4 time slots). 4 ERA5-Land ZIP files per date.

S1.7 Refractivity Computation

Applied at each ERA5 model level k for each of the three spatial sampling points (A, M, B):

Step 1 — Pressure: $p(k) = [a(k) + b(k) \times p_{\text{surface}}] / 100$ [hPa]

Step 2 — Geometric height: $z(k) = \phi(k) / g_0$, $g_0 = 9.80665$ m/s²

Step 3 — Water vapour: $e(k) = q(k) \times p(k) / (0.622 + 0.378 \times q(k))$ [hPa]

Step 4 — Refractivity (Smith-Weintraub, 1953): $N(k) = 77.6 \times p(k)/T(k) + 373000 \times e(k)/T(k)^2$

Real example — Riyadh, 15 July 2025, 10:00 UTC, observer (24.60N, 46.80E, 600m): T=311.51K, RH=8.1%, e=5.404hPa, p=943.5hPa. Dry term=235.07 (92%). Wet term=20.74 (8%). Total N=255.81.

Step 5 — dN/dz by least-squares regression across model levels within observer-target altitude range.

Step 6 — $\kappa = -(R \times dN/dz \times 10^{-6})$, $R = 6,371,000$ m

Step 7 — Three-point ensemble: $\kappa_{path} = (\kappa_A + \kappa_M + \kappa_B) / 3$

Step 8 — Linear temporal interpolation between bracketing 6-hourly ERA5 analyses.

S2. Three-Point Spatial Sampling — Full Dataset (41 Paths)

S2.1 Methodology

Three-point spatial sampling (kappa_3pt) was compared against single midpoint sampling (kappa_mid) using the dedicated /api/compute/compare endpoint deployed on KAPPA Arc V3 backend. Both values are derived from the same ERA5 L137 data pipeline — the only difference is whether the path-integrated refractivity gradient uses profiles at observer (A), midpoint (M), and target (B) ensemble-averaged (three-point), or only the midpoint profile (M) alone. This design isolates the spatial sampling methodology as the sole variable.

API endpoint: kappa-backend-v3-production.up.railway.app/api/compute/compare. All paths computed at 10:00 UTC. Summer paths dated 15 July 2025; winter paths (CAU-1, AFR-3, ZAF-1) dated 15 January 2025. Path lengths range from 1.7 km (MED-1, Cortes de Pallás baseline, retained from main paper validation) to 235 km (USA-2, Houston to Austin), with the majority of paths in the 35–90 km range representing typical geodetic line-of-sight survey distances.

S2.2 Complete Results — 41 Paths

Path	Region	Observer (lat, lon, elev)	Target (lat, lon, elev)	dist (km)	kappa_3pt	kappa_mid	Delta-k	Direction
SA-1	Saudi Arabia	24.60N,46.80E,600 m	24.95N,47.20E,650 m	56.1	0.1909	0.1912	- 0.0003	mid>3pt
SA-2	Saudi Arabia	26.42N,49.98E,10 m	26.80N,50.35E,80m	56.0	0.2359	0.2329	+0.003 0	3pt>mid
SA-3	Saudi Arabia	21.48N,39.19E,15 m	21.85N,39.55E,380 m	55.5	0.2862	0.2983	- 0.0121	mid>3pt
MED-1	Spain	39.220N,1.175W,4 20m	39.232N,1.162W,6 80m	1.7	0.2136	0.2136	+0.000 0	equal
MED-2	Spain	41.38N,2.17E,10m	41.65N,2.45E,250m	38.0	0.2682	0.2678	+0.000 5	3pt>mid
MED-3	France	45.75N,4.85E,170 m	45.18N,5.72E,212m	92.8	0.2502	0.2500	+0.000 3	3pt>mid
CAU-1	Caucasus	43.05N,44.68E,270 0m	42.44N,44.05E,519 3m	85.1	0.1657	0.1679	- 0.0022	mid>3pt
TUR-1	Turkey	39.92N,32.85E,890 m	40.20N,33.10E,950 m	37.7	0.2163	0.2155	+0.000 8	3pt>mid
TUR-2	Turkey	36.90N,30.70E,200 m	37.20N,30.55E,180 0m	35.9	0.2450	0.2448	+0.000 2	3pt>mid
USA-1	USA	34.05N,118.24W,7	34.05N,117.75W,2	45.1	0.2777	0.2697	+0.008	3pt>mid

		0m	85m				0	
USA-2	USA	29.76N,95.37W,15 m	30.27N,97.74W,15 0m	235.1	0.3016	0.3075	- 0.0059	mid>3pt
USA-3	USA	32.78N,96.80W,13 9m	33.15N,96.45W,17 5m	52.5	0.3087	0.3110	- 0.0023	mid>3pt
AFR-1	Egypt	30.06N,31.25E,75 m	30.40N,31.55E,100 m	47.5	0.2380	0.2371	+0.000 9	3pt>mid
AFR-2	Kenya	1.29S,36.82E,1680 m	1.00S,37.10E,1750 m	44.8	0.2159	0.2153	+0.000 5	3pt>mid
AFR-3	S.Africa	26.20S,28.04E,175 0m	25.85S,28.35E,168 0m	49.7	0.2393	0.2435	- 0.0042	mid>3pt
IND-1	India	18.52N,73.86E,560 m	18.85N,74.15E,720 m	47.7	0.2675	0.2662	+0.001 3	3pt>mid
IND-2	India	28.61N,77.21E,220 m	28.95N,77.55E,260 m	50.3	0.2869	0.2877	- 0.0008	mid>3pt
IND-3	India	12.97N,77.59E,920 m	13.30N,77.85E,870 m	46.3	0.2436	0.2432	+0.000 4	3pt>mid
CHN-1	China	39.90N,116.40E,50 m	40.20N,116.70E,80 m	42.0	0.2727	0.2717	+0.001 0	3pt>mid
CHN-2	China	31.23N,121.47E,5 m	31.55N,121.20E,30 m	43.9	0.2848	0.2710	+0.013 8	3pt>mid
CHN-3	China	30.57N,104.07E,50 0m	30.90N,104.40E,52 0m	48.4	0.2739	0.2720	+0.001 9	3pt>mid
SCAN- 1	Norway	59.91N,10.75E,23 m	60.20N,11.05E,200 m	36.3	0.2574	0.2582	- 0.0008	mid>3pt
CAS-1	Kazakhsta n	43.25N,76.95E,773 m	43.55N,77.25E,850 m	41.2	0.2185	0.2138	+0.004 6	3pt>mid
SEA-1	Vietnam	21.03N,105.85E,15 m	21.35N,106.10E,50 m	44.0	0.2940	0.2937	+0.000 3	3pt>mid
ARG-1	Argentina	34.60S,58.38W,25 m	34.25S,58.00W,60 m	52.2	0.2548	0.2513	+0.003 5	3pt>mid
NGA-1	Nigeria	9.07N,7.40E,840m	9.40N,7.70E,900m	49.3	0.2455	0.2464	- 0.0009	mid>3pt
IRN-1	Iran	35.69N,51.39E,119 0m	36.00N,51.70E,135 0m	44.4	0.2050	0.2012	+0.003 8	3pt>mid
JPN-1	Japan	35.68N,139.69E,40 m	35.40N,140.00E,80 m	41.9	0.2966	0.2971	- 0.0005	mid>3pt
POL-1	Poland	52.23N,21.01E,110 m	52.55N,21.35E,150 m	42.4	0.2512	0.2519	- 0.0007	mid>3pt
THA-1	Thailand	13.75N,100.52E,15 m	14.10N,100.85E,50 m	52.8	0.2845	0.2837	+0.000 8	3pt>mid
KOR-1	South Korea	37.56N,127.00E,50 m	37.85N,127.30E,10 0m	41.7	0.2868	0.2864	+0.000 4	3pt>mid
VNM-	Vietnam	10.82N,106.66E,10	11.15N,106.95E,50	48.5	0.2847	0.2840	+0.000	3pt>mid

1		m	m				7	
EGY-1	Egypt	24.09N,32.90E,95 m	24.45N,33.20E,120 m	50.3	0.2260	0.2263	- 0.0003	mid>3pt
MAR-1	Morocco	33.57N,7.59W,56m	33.90N,7.25W,200 m	48.3	0.2783	0.2800	- 0.0017	mid>3pt
GBR-1	UK	51.51N,0.13W,20m	51.80N,0.20E,80m	39.5	0.2437	0.2425	+0.001 1	3pt>mid
AUS-1	Australia	33.87S,151.21E,50 m	33.50S,151.00E,18 0m	45.5	0.2328	0.2312	+0.001 6	3pt>mid
CAN-1	Canada	43.65N,79.38W,90 m	43.95N,79.05W,25 0m	42.6	0.2501	0.2476	+0.002 5	3pt>mid
ZAF-1	South Africa	33.93S,18.42E,20m	33.60S,18.75E,300 m	47.7	0.2256	0.2258	- 0.0001	mid>3pt
MEX-1	Mexico	19.43N,99.13W,22 40m	19.75N,98.80W,24 00m	49.6	0.2203	0.2153	+0.005 0	3pt>mid
PAK-1	Pakistan	24.86N,67.01E,10 m	25.20N,67.35E,100 m	51.0	0.2786	0.2777	+0.000 9	3pt>mid
RUS-1	Russia	55.75N,37.62E,156 m	56.05N,37.95E,200 m	39.2	0.2510	0.2508	+0.000 1	3pt>mid

Table S6. Complete 41-path three-point vs single midpoint spatial sampling comparison. All kappa values computed from the same ERA5 L137 data pipeline using /api/compute/compare endpoint. kappa_3pt = ensemble average of profiles at A, M, B. kappa_mid = midpoint profile only. Path lengths 35-90 km except MED-1 (1.7 km, Cortes de Pallas baseline retained from main paper validation) and USA-2 (235 km). All computations at 10:00 UTC; winter dates for CAU-1, AFR-3, ZAF-1 (15 January 2025); all others 15 July 2025.

S2.3 Statistical Summary — 41 Paths

Statistic	Value
Total paths	41
3pt > midpoint	26 of 41 (63.4%)
midpoint > 3pt	14 of 41 (34.1%)
Equal	1 of 41 (2.4%) — MED-1, Cortes de Pallas 1.7 km baseline
Mean Delta-kappa	+0.0006
Standard deviation	0.0037
Minimum Delta-kappa	-0.0121 (SA-3, Jeddah inland)
Maximum Delta-kappa	+0.0138 (CHN-2, Shanghai inland)
Median Delta-kappa	+0.0004
Path length range	1.7 to 235 km (median 47 km)
Geographic coverage	24 countries across all inhabited continents
Data source	ERA5 L137 via /api/compute/compare endpoint — same pipeline for both methods

Table S7. Statistical summary — three-point vs single midpoint spatial sampling, 41 paths. All values from real API responses using same ERA5 L137 data for both methods.

S2.4 Scientific Interpretation

Three-point spatial sampling is physically more representative than single midpoint sampling for all observation paths. Single midpoint sampling assumes that atmospheric conditions at the geometric centre of the path represent conditions along the entire path — an approximation that becomes less valid as horizontal atmospheric gradients increase. Three-point sampling explicitly characterises conditions at both endpoints and the midpoint, providing a more accurate path-integrated estimate regardless of whether the resulting kappa is higher or lower than the midpoint-only value.

The 14 cases where single midpoint produced higher kappa are not evidence of midpoint superiority — they reflect natural horizontal atmospheric variability where the midpoint profile happens to be locally more refractive than the endpoint average on that specific day. The methodology advantage becomes numerically significant for paths crossing horizontal atmospheric gradients such as coastal-to-inland transitions, orographic features, and thermal regime boundaries. The small mean difference (+0.0006) for 35-90 km paths reflects ERA5's 0.25 degree horizontal resolution, within which three sampling points on a path of this length typically fall within the same or adjacent grid cells.

S3. Global Radiosonde vs ERA5 Surface Temperature — Full Dataset

S3.1 Methodology

ERA5 2m temperature was obtained from KAPPA V3 surface composite mode (/api/compute/surface) at the exact station coordinates and at the same UTC hour as the radiosonde launch — temporal and spatial matching essential for valid comparison. An earlier version of this comparison using a fixed 12:00 UTC ERA5 request showed substantially larger apparent errors for several stations (Cairo July: 6.92 °C mismatched vs 0.09 °C matched; Beijing January: 8.10 °C mismatched vs 1.04 °C matched), confirming that temporal matching is essential.

Request parameters: lat_a = station latitude, lon_a = station longitude, elev_a = 10m, lat_b = lat_a + 0.01, lon_b = lon_a + 0.01, elev_b = 10m, time = sonde UTC launch hour, mode = survey. Radiosonde surface temperature from NOAA IGRA2 pre-processed archive in Amazon S3. Surface level = highest p_hpa. Temperature field: temp_c (degrees Celsius). Dates: 15 July 2025 and 15 January 2025. Auckland provides southern hemisphere maritime coverage.

S3.2 Complete Comparison Results

Station ID	Location	Country	Lat, Lon	Sonde UTC	Jul T_sonde	Jul T_E RA5	Jul DT	Sonde UTC	Jan T_sonde	Jan T_E RA5	Jan DT
SAM00040417	Dammam	Saudi Arabia	26.45N,49.81E	1200	47.0	40.8	6.25	1200	21.6	17.7	3.92
SAM00040373	Riyadh proxy	Saudi Arabia	24.93N,46.72E	0000	34.8	36.8	1.98	1200	19.0	15.5	3.54
AEM00041217	Dubai	UAE	25.25N,55.33E	1200	41.0	39.4	1.62	0000	16.6	18.3	1.73
EGM00062306	Cairo	Egypt	30.08N,31.28E	0000	23.2	23.1	0.09	0000	12.2	16.8	4.56
KEM00063741	Nairobi	Kenya	1.32S,36.92E	0000	14.6	13.1	1.49	0000	15.0	16.1	1.06
INM00043003	Mumbai	India	19.12N,72.85E	1200	28.8	24.6	4.24	1200	27.6	23.4	4.21
INM00042182	Delhi	India	28.58N,77.20E	0000	25.4	23.9	1.45	1200	15.8	14.5	1.27
SPM00008221	Madrid	Spain	40.45N,3.55W	0000	27.0	27.7	0.73	0000	0.4	6.6	6.20
FRM000075	Bordeau	France	44.83N,0.7	0000	19.2	17.7	1.51	1200	2.4	2.2	0.22

10	x		0W								
GMM00010771	Munich	Germany	48.25N,11.55E	1200	22.6	17.6	4.99	0000	-2.3	0.1	2.36
ITM00016245	Milan	Italy	45.43N,9.28E	1200	30.8	26.5	4.29	1200	8.0	3.8	4.17
RSM00027730	Moscow	Russia	55.75N,37.57E	0000	18.4	21.2	2.79	1200	0.4	-1.4	1.78
CHM00054511	Beijing	China	39.93N,116.28E	1200	22.5	20.5	2.03	0000	-7.0	-6.0	1.04
BRM00083779	Sao Paulo	Brazil	23.20S,45.87W	0000	15.4	14.1	1.26	1200	21.2	19.8	1.44
USM00072501	New York	USA	40.78N,73.90W	0000	25.5	22.6	2.91	0000	-2.2	-3.5	1.35
NZM00093112	Auckland	NZ	37.00S,174.80E	0000	12.4	10.3	2.12	1200	16.8	14.9	1.90

Table S8. Complete ERA5 vs IGRA2 radiosonde surface temperature comparison. ERA5 at exact station coordinates and matching UTC sonde hour. 16 stations, 32 valid comparisons.

S3.3 Melbourne — ERA5 Limitation Note

Date	Season	T_sonde (C)	T_ERA5 (C)	ΔT (C)	Notes
January 2025	Australian summer	31.4	17.0	14.37	ERA5 severe underestimation
July 2025	Australian winter	13.3	6.2	7.13	ERA5 underestimation continues
August 2025	Australian winter	13.0	9.19	3.81	Smallest gap — stable winter

Table S9. Melbourne (ASM00094326) ERA5 vs radiosonde. ERA5 consistently underestimates surface temperature due to unresolvable land-sea interaction in Port Phillip Bay at ERA5 0.25° resolution. Not included in the 16-station analysis.

S3.4 Statistical Summary — 32 Valid Comparisons

Statistic	Value
Stations	16
Dates per station	2 (15 July 2025 + 15 January 2025)
Valid comparisons	32
Methodology	ERA5 at exact station coordinates, matching UTC sonde hour
Mean absolute ΔT	2.52 °C
Standard deviation	1.61 °C
Minimum ΔT	0.09 °C (Cairo, July 2025, 00:00 UTC)

Maximum $ \Delta T $	6.25 °C (Dammam, July 2025, 12:00 UTC)
Best agreement	Flat/maritime: Bordeaux 0.22 °C; Nairobi 1.06 °C; Delhi 1.27 °C
Largest departures	Dammam Jul 6.25 °C (desert heating); Madrid Jan 6.20 °C (inversion)
kappa uncertainty equivalent	approx. ± 0.004

Table S10. Statistical summary — ERA5 vs IGR2 radiosonde, 32 valid comparisons, 16 stations. All values from real data with matched time and location methodology.

S4. KAPPA Arc V3 System Parameters

S4.1 API Endpoints Used in This Study

Endpoint	URL	Used For
ERA5 L137 LOS mode	kappa-backend-v3-production.up.railway.app/api/compute/elite	Three-point spatial sampling — ERA5 137-level reanalysis
Surface (Levelling/Survey)	kappa-backend-v3-production.up.railway.app/api/compute/surface	Saudi validation; radiosonde comparison
Compare (3pt vs mid)	kappa-backend-v3-production.up.railway.app/api/compute/compare	Three-point vs midpoint evaluation (41 paths)
/api/compute/elite/dual	kappa-backend-v3-production.up.railway.app/api/compute/elite/dual	Dual gradient computation: OLS + numerical integration. Returns delta_k, agreement, recommendation.
/api/ai/interpret	kappa-backend-v3-production.up.railway.app/api/ai/interpret	AI interpretation via Claude Sonnet 4 (Anthropic). Input: result, mode, question. Avg response: 2-4 sec.

Table S11. API endpoints used in this study.

S4.2 Automated Daily Data Update Schedule

UTC Time	Script	Data Product
03:00	kappa_radiosonde_download.py	NOAA IGRA2 — regional Saudi/Gulf stations
04:00	kappa_radiosonde_global_download.py	NOAA IGRA2 — 889 global stations
06:00	kappa_download.py	ERA5 pressure-level reanalysis (37 pressure levels)
07:00	kappa_l137_download.py	ERA5 L137 137-level reanalysis + regional extraction
09:00	kappa_metar_download.py	METAR surface aviation observations
10:00	kappa_gfs_download.py	GFS numerical weather prediction (0.25 deg)
14:00	kappa_era5_land_download.py	ERA5-Land (skin T, 2m T, dewpoint, surface pressure)
15:00	kappa_era5_single_download.py	ERA5-Single (boundary layer height, TCWV, SST)

Table S12. Automated daily data update schedule. All scripts on AWS EC2 (m7i-flex.large, eu-north-1).

S4.3 Computational Performance

Metric	Value	Basis
Cold request time (mean)	17.3 s	120 test computations, all 8 modes, 6 regions

Cold request time (range)	11-23 s	Min and max across 120 computations
Cached request time (mean)	1.4 s	Redis cache hit
Cached request time (range)	0.8-2.1 s	Min and max cached responses
Global ERA5 L137 file size	~192 MB	Full global NetCDF per time slot
Regional extracted file size	7-66 MB	Depends on geographic region
Cache TTL	24 hours	Redis time-to-live
Ray geometry points	1,000	Equally-spaced along observer-target chord
Binary search iterations	up to 50	Convergence to machine precision ($\sim 10^{-15}$)

Table S13. Measured computational performance metrics.

S4.4 Complete API Response Structure

Field	Type	Description
k	float	Atmospheric refraction coefficient
k_required	float	Minimum kappa for geometric visibility (binary search)
deficit	float	k minus k_required; positive = visible
verdict	string	visible or not_visible
dndh	float	Refractivity gradient dN/dh in N/km
n_avg	float	Mean refractivity along path
source	string	Data source used
layers	array	Atmospheric profile: p, alt, T, Tc, RH, e, N per level
sliced_layers	array	Profile subset within observer-target altitude range
corrections.cr_m	float	C+R correction in metres for the observation path
corrections.angular_arcsec	float	Angular refraction in arcseconds
inversion.detected	boolean	True if thermal inversion detected
inversion.classification	string	Inversion classification or none
ducting.detected	boolean	True if radio ducting detected
ducting.trapping_layer_alt_m	float/null	Altitude of ducting layer if detected
ducting.m_profile	array	Modified refractivity $M = N + z/R \times 10^6$ per level
ray.visible	boolean	True if min ray clearance ≥ 0 at all 1,000 positions
ray.h_min_m	float	Minimum ray height above Earth surface (m)
ray.h_mid_m	float	Ray height at geometric midpoint (m)
ray.sagitta_m	float	Total ray sag from straight-line path (m)
ray.points	array	1,000 ray heights at equally-spaced path positions (m)

radiosonde.station_id	string	IGRA2 nearest station identifier
radiosonde.station_name	string	Station name
radiosonde.distance_km	float	Distance from path midpoint to station (km)
radiosonde.levels	array	Sounding levels: p_hpa, alt_m, temp_c, dewp_c
date	string	Requested date (YYYY-MM-DD)
time	string	Requested time (HH:MM UTC)
dist_km	float	Observer-to-target distance (km)
elev_a / elev_b	float	Observer / target elevation (m)

Table S14. Complete KAPPA Arc V3 API response structure.

S4.5 Data Citations

ERA5: Hersbach, H., et al. (2020). QJRMS, 146(730), 1999-2049. <https://doi.org/10.1002/qj.3803>

ERA5-Land: Muñoz-Sabater, J., et al. (2021). ESSD, 13(9), 4349-4383. <https://doi.org/10.5194/essd-13-4349-2021>

IGRA2: NOAA NCEI. (2023). Integrated Global Radiosonde Archive, Version 2.2. <https://doi.org/10.7289/V5X63K0Q>

Refractivity: Smith, E.K., & Weintraub, S. (1953). IRE Proceedings, 41(8), 1035-1037.

Saudi field data: General Commission for Survey. (2018). RCD_PLO Project Final Report.

Spain field data: Lujan, R., et al. (2025). Sensors, 25(7), 1981. <https://doi.org/10.3390/s25071981>

S5. Dual Gradient Computation — Complete Dataset

This section provides the complete dual gradient computation dataset for all 41 globally distributed validation paths. OLS regression and trapezoidal numerical integration were applied to ERA5 L137 data at 10:00 UTC. API endpoint: </api/compute/elite/dual>.

S5.1 Complete Results — 41 Paths

Path ID	Description	OLS κ	Num. κ	$\Delta\kappa$	Agreement	Layers	Date
SA-1	Riyadh corridor, Saudi Arabia	0.1909	0.1909	0.0000	Excellent	2	2025-07-15
SA-2	Dammam inland, Saudi	0.2359	0.2359	0.0000	Excellent	2	2025-07-15

	Arabia						
SA-3	Jeddah inland, Saudi Arabia	0.2862	0.2848	-0.0014	Good	3	2025-07-15
MED-1	Cortes de Pallás, Spain	0.2136	0.2136	0.0000	Excellent	2	2025-07-15
MED-2	Barcelona inland, Spain	0.2682	0.2682	0.0000	Excellent	2	2025-07-15
MED-3	Lyon to Grenoble, France	0.2502	0.2502	0.0000	Excellent	2	2025-07-15
CAU-1	Caucasus baseline, Georgia	0.1657	0.1654	-0.0003	Excellent	7	2025-01-15
TUR-1	Ankara inland, Turkey	0.2163	0.2163	0.0000	Excellent	2	2025-07-15
TUR-2	Taurus inland, Turkey	0.2450	0.2457	+0.0007	Excellent	6	2025-07-15
USA-1	Los Angeles, USA	0.2777	0.2777	0.0000	Excellent	2	2025-07-15
USA-2	Houston to Austin, USA	0.3016	0.3017	+0.0001	Excellent	3	2025-07-15
USA-3	Dallas plain, USA	0.3087	0.3087	0.0000	Excellent	2	2025-07-15
AFR-1	Cairo inland, Egypt	0.2380	0.2380	0.0000	Excellent	2	2025-07-15
AFR-2	Nairobi inland, Kenya	0.2159	0.2159	0.0000	Excellent	2	2025-07-15
AFR-3	Johannesburg, South Africa	0.2393	0.2393	0.0000	Excellent	2	2025-01-15
IND-1	Pune plateau, India	0.2675	0.2675	0.0000	Excellent	2	2025-07-15
IND-2	Delhi inland, India	0.2869	0.2869	0.0000	Excellent	2	2025-07-15
IND-3	Bangalore plateau, India	0.2436	0.2436	0.0000	Excellent	2	2025-07-15
CHN-1	Beijing	0.2727	0.2727	0.0000	Excellent	2	2025-07-15

	plain, China						
CHN-2	Shanghai inland, China	0.2848	0.2848	0.0000	Excellent	2	2025-07-15
CHN-3	Chengdu basin, China	0.2739	0.2739	0.0000	Excellent	2	2025-07-15
SCAN-1	Oslo inland, Norway	0.2574	0.2574	0.0000	Excellent	2	2025-07-15
CAS-1	Almaty valley, Kazakhstan	0.2185	0.2208	+0.0023	Good	3	2025-07-15
SEA-1	Hanoi inland, Vietnam	0.2940	0.2940	0.0000	Excellent	2	2025-07-15
ARG-1	Buenos Aires, Argentina	0.2548	0.2548	0.0000	Excellent	2	2025-07-15
NGA-1	Abuja inland, Nigeria	0.2455	0.2455	0.0000	Excellent	2	2025-07-15
IRN-1	Tehran inland, Iran	0.2050	0.2050	0.0000	Excellent	2	2025-07-15
JPN-1	Tokyo inland, Japan	0.2966	0.2966	0.0000	Excellent	3	2025-07-15
POL-1	Warsaw inland, Poland	0.2512	0.2512	0.0000	Excellent	2	2025-07-15
THA-1	Bangkok inland, Thailand	0.2845	0.2845	0.0000	Excellent	2	2025-07-15
KOR-1	Seoul inland, South Korea	0.2868	0.2866	-0.0002	Excellent	3	2025-07-15
VNM-1	Ho Chi Minh City, Vietnam	0.2847	0.2847	0.0000	Excellent	2	2025-07-15
EGY-1	Aswan inland, Egypt	0.2260	0.2260	0.0000	Excellent	2	2025-07-15
MAR-1	Casablanca inland, Morocco	0.2783	0.2783	0.0000	Excellent	2	2025-07-15
GBR-1	London inland, UK	0.2437	0.2437	0.0000	Excellent	2	2025-07-15
AUS-1	Sydney inland, Australia	0.2328	0.2328	0.0000	Excellent	2	2025-07-15

CAN-1	Toronto inland, Canada	0.2501	0.2501	0.0000	Excellent	2	2025-07-15
ZAF-1	Cape Town, South Africa	0.2256	0.2256	0.0000	Excellent	2	2025-01-15
MEX-1	Mexico City, Mexico	0.2203	0.2195	-0.0008	Excellent	3	2025-07-15
PAK-1	Karachi inland, Pakistan	0.2786	0.2786	0.0000	Excellent	2	2025-07-15
RUS-1	Moscow inland, Russia	0.2510	0.2510	0.0000	Excellent	2	2025-07-15

Table S15. Complete dual gradient computation results — 41 globally distributed paths. Agreement: *EXCELLENT* $|\Delta\kappa| < 0.001$; *GOOD* 0.001–0.005; *DIVERGENT* ≥ 0.005 .

S5.2 Statistical Summary

Metric	Value	Notes
EXCELLENT ($ \Delta\kappa < 0.001$)	39/41 (95.1%)	OLS and numerical equivalent
GOOD (0.001–0.005)	2/41 (4.9%)	SA-3 Jeddah, CAS-1 Almaty; 3 ERA5 levels each
DIVERGENT (≥ 0.005)	0/41 (0.0%)	No divergent cases observed
Mean $ \Delta\kappa $	0.000126	All 41 paths
Max $ \Delta\kappa $	0.002287	CAS-1, Almaty valley, Kazakhstan
Min $ \Delta\kappa $	0.000000	29 paths with 2 ERA5 levels (mathematically equivalent)
2-level paths	29/41 (70.7%)	$\Delta\kappa = 0.0000$ — mathematical equivalence for 2-point profiles
3+ level paths	12/41 (29.3%)	Non-linearity detectable; numerical integration adds value

Table S16. Agreement classification and statistical summary — dual gradient computation, 41 paths.

S6. AI Interpretation Layer — Full Examples and Accuracy Verification

All responses generated by Claude Sonnet 4 (Anthropic, 2025) from real KAPPA Arc V3 computation results. API endpoint: /api/ai/interpret.

S6.1 Complete AI Outputs — All Eight Modes

Mode	KAPPA Result (key values)	AI Interpretation (verbatim excerpt)
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LOS Long Range	$\kappa=0.1880$, dist=56.18km, $\kappa_{\text{req}}=0.000$, deficit=+0.188	Computed κ (0.188) far exceeds required κ (0.000) — target is visible with large margin. Refraction 45% stronger than standard. ACTION: Proceed with optical observation or link planning.
Survey/EDM	$\kappa=0.1998$, dist=2.45km, C+R=0.283m	Correct C+R = 0.283 m. Standard applies 0.349 m — standard overcorrects by 66 mm (KAPPA gives the smaller, physically correct correction). ACTION: Enter $\kappa = 0.1998$ into total station atmospheric settings before measuring.
Levelling	$\kappa=0.1998$, dist=0.6km, C+R=0.017m	Correct C+R = 17 mm per sight. Standard applies 21 mm — error of -4 mm per setup. Over a 100 km traverse (~167 setups), cumulative error ≈ 668 mm, exceeding first-order tolerance. ACTION: Enter $\kappa = 0.20$ before fieldwork.
Marine	$\kappa=0.081$, dist=59.85km, BLOCKED	Target blocked — $\kappa = 0.081$ far below required $\kappa = 0.629$. Low κ over warm water is physically correct. ACTION: Use GPS rather than optical LOS at this range.
Drone/UAV	$\kappa=0.1965$, dist=10.32km, GFS	Stable atmosphere. No inversion detected. Good conditions for UAV survey. ACTION: Safe to fly — normal operations.
Astronomy	$\kappa=0.1883$, dist=56.18km, 22:00 UTC	Angular refraction = 170.9 arcseconds upward. No inversion — good seeing. ACTION: Enter $\kappa = 0.1883$ into atmospheric refraction model before pointing.
Radar/Micro	$\kappa=0.1999$, dist=56.18km, no ducting	Normal propagation. No ducting risk. ACTION: Use $\kappa = 0.20$ in path profile software; +3 dB fade margin recommended.
Tunnel	$\kappa=0.0716$, dist=2.1km, C+R=0.297m	Correct C+R = 297 mm. Standard applies 256 mm — +41 mm error. Lower κ = LARGER C+R (counter-intuitive). Exceeds ± 25 mm breakthrough tolerance. ACTION: Enter $\kappa = 0.07$ before portal measurements.

Table S17. AI interpretation layer outputs for all eight operational modes.

S6.2 Follow-Up Question Examples

Mode	Follow-up Question	AI Response (excerpt)
Levelling	What if I have 50 setups today?	50 setups $\times$ 4 mm/setup = 200 mm

		cumulative error. Enter $\kappa = 0.20$ before starting.
Levelling	Is morning better than afternoon?	Morning (04:00–09:00 UTC) significantly better. Afternoon κ drops to 0.08–0.12 in desert. ERA5 shows 47°C surface by midday. Complete all levelling before 09:00 UTC.
LOS	What focal length do I need?	For 56 km path, 1000–1200 mm recommended for fine detail. Shimmer limits effective resolution at this distance.
Tunnel	What if I use standard correction?	Standard applies 256 mm instead of 297 mm — +41 mm error at breakthrough. Exceeds ± 25 mm tolerance. Requires remediation.

Table S17b. Follow-up question capability verification.

S6.3 Accuracy Verification

Mode	Parameter	KAPPA Value	AI Stated	Match?
LOS	κ computed	0.1880	0.188	Yes
LOS	κ required	0.000	0.000	Yes
Survey	κ computed	0.1998	0.1998	Yes
Survey	KAPPA C+R	0.283 m	0.283 m	Yes
Survey	Standard C+R	0.349 m	0.349 m	Yes
Survey	Correction error	–66 mm	–66 mm	Yes
Levelling	C+R per sight	17 mm	17 mm	Yes
Levelling	Standard C+R	21 mm	21 mm	Yes
Levelling	Error per setup	–4 mm	–4 mm	Yes
Levelling	Cumulative 100 km	668 mm	668 mm	Yes
Levelling	1st-order tolerance	12.6 mm	12.6 mm	Yes
Tunnel	KAPPA C+R	297 mm	297 mm	Yes
Tunnel	Standard C+R	256 mm	256 mm	Yes
Tunnel	Correction error	+41 mm	+41 mm	Yes
Astronomy	Angular correction	170.9"	170.9"	Yes
Levelling FU	50 setups cumul.	200 mm	200 mm	Yes
Levelling FU	1st-order 10km	12.6 mm	12.6 mm	Yes

Table S18. AI accuracy verification — 17 tested parameters, all match KAPPA-computed values. No hallucination detected.

S7. Tropical Radiosonde κ Validation — Full Dataset

S7.1 Station Details and IGRA2 Identifiers

Table S19 provides full station details for the four ICAO-standard radiosonde stations used in the tropical κ validation (Section 4.4 of the main paper). All four stations report significant atmospheric levels with altitude and dewpoint at every level — the density criterion required for κ_{sonde} computation over the 0–500 m analysis window. Eight candidate stations that failed the density criterion (reporting standard pressure levels only) are listed in Table S20.

Station	IGRA2 ID	WMO #	Lat	Lon	Elev (m)	Ocean Basin	Operator
Grantley Adams, Barbados	BBM00078954	78954	13.07° N	59.49° W	57	Atlantic	US NWS/Caribbean
Pohnpei Intl, Micronesia	FMM00091334	91334	6.96° N	158.21° E	6	W. Pacific	US NWS Pacific
Luis Muñoz Marín, San Juan PR	RQM00078526	78526	18.43° N	66.00° W	3	Caribbean	US NWS
Singapore/Changi Airport	SNM00048698	48698	1.37° N	103.98° E	16	SE Asian	Meteorological Service Singapore

Table S19. IGRA2 station details for tropical κ validation. All stations report significant atmospheric levels with altitude and dewpoint at full vertical resolution (6–8 usable levels in the 0–500 m analysis window). IGRA2 data retrieved from NOAA NCEI (<https://www.ncei.noaa.gov/data/integrated-global-radiosonde-archive/>). RQM prefix = Puerto Rico ICAO territory code (not USM as might be assumed).

Station	ID attempted	File size	Failure reason	Last data
Guam/Tiyan	GUM00091212	196 bytes	Invalid ID — 404	N/A
San Juan (US prefix)	USM00078526	196 bytes	Invalid ID — use RQM	N/A
Diego Garcia	DGM00061967	196 bytes	Invalid ID — 404	N/A
Trinidad/Piarco	TTM00078970	196 bytes	Invalid ID — 404	N/A
Male, Maldives	MVM00043599	5.9 MB	Stopped Dec 2021	Dec 2021
Accra, Ghana	GHM00065418	633 KB	Stopped 2008	Sep 2008
Manaus, Brazil	BRM00082332	20.4 MB	Sparse: standard levels only (2 levels in 0–900m)	Active
Bangkok	THM00048453	—	Sparse: standard levels only	Active

Table S20. Candidate tropical stations that failed the density criterion. '196 bytes' = HTTP 404 response — station ID not found in IGRA2 archive. Manaus and Bangkok are active stations but report only mandatory standard pressure levels, providing ≤ 2 altitude-tagged levels in the 0–500 m window.

S7.2 Sounding Level Details

Table S21 provides the raw level-by-level data used for κ _sonde computation at each station and date. Levels are filtered to those within the 0–500 m analysis window with both altitude and dewpoint depression reported. N (refractivity) and wet% are computed from the Smith-Weintraub formula at each level.

Station	Date	Alt (m)	P (hPa)	T (°C)	Td (°C)	N	Wet%
Barbados	Jul 2025	57	1010.7	27.3	23.5	374.1	30.2
Barbados	Jul 2025	152	1000.0	26.6	23.2	369.6	29.9
Barbados	Jul 2025	221	992.2	25.9	22.8	366.0	30.1
Barbados	Jul 2025	300	983.4	25.2	22.4	362.4	30.0
Barbados	Jul 2025	444	967.4	23.9	21.7	356.1	29.8
Barbados	Jul 2025	600	950.3	23.1	19.9	347.3	28.5
Barbados	Jul 2025	836	925.0	21.3	19.2	339.0	28.6
Barbados	Jan 2025	57	1008.1	25.6	20.8	366.9	28.0
Barbados	Jan 2025	108	1002.5	25.6	20.6	364.4	27.8
Barbados	Jan 2025	128	1000.0	25.4	20.5	363.0	27.8
Barbados	Jan 2025	207	991.0	24.7	20.3	358.5	27.8
Barbados	Jan 2025	300	980.7	23.7	20.0	354.0	27.9
Barbados	Jan 2025	528	955.4	21.7	19.1	343.2	27.6
Barbados	Jan 2025	600	947.5	21.5	17.3	333.2	25.8
Barbados	Jan 2025	809	925.0	20.1	15.2	320.8	24.2
Pohnpei	Jul 2025	6	1009.3	29.4	24.4	383.4	32.5
Pohnpei	Jul 2025	88	1000.0	27.9	23.5	376.9	32.2
Pohnpei	Jul 2025	132	995.0	27.5	23.2	374.1	31.9
Pohnpei	Jul 2025	140	994.1	27.4	23.1	373.4	31.8
Pohnpei	Jul 2025	625	941.1	23.2	21.4	354.6	31.3
Pohnpei	Jul 2025	775	922.5	22.2	20.5	346.1	30.8
Pohnpei	Jan 2025	6	1010.5	29.6	25.9	395.0	34.4
Pohnpei	Jan 2025	73	1002.9	28.4	22.7	371.2	29.9
Pohnpei	Jan 2025	99	1000.0	28.1	22.5	369.6	29.8
Pohnpei	Jan 2025	144	994.9	27.6	22.5	369.1	30.2

Pohnpei	Jan 2025	763	927.4	21.7	21.2	352.0	30.9
Pohnpei	Jan 2025	786	925.0	21.6	20.5	347.0	30.3
San Juan	Jul 2025	3	1017.6	27.8	21.1	365.4	28.2
San Juan	Jul 2025	158	1000.0	27.1	20.8	360.0	27.9
San Juan	Jul 2025	307	983.3	25.8	20.3	354.6	27.9
San Juan	Jul 2025	357	977.7	25.3	20.1	352.7	27.9
San Juan	Jul 2025	389	974.2	25.4	18.8	344.0	26.3
San Juan	Jul 2025	592	951.9	24.1	17.1	330.8	24.9
San Juan	Jul 2025	843	925.0	21.7	17.3	328.1	25.6
San Juan	Jul 2025	876	921.4	21.5	16.8	324.8	25.4
San Juan	Jan 2025	3	1017.0	25.0	19.4	359.2	25.9
San Juan	Jan 2025	34	1013.4	25.1	18.6	353.5	25.3
San Juan	Jan 2025	151	1000.0	24.1	18.0	348.1	25.2
San Juan	Jan 2025	792	928.9	18.1	16.3	328.9	25.2
San Juan	Jan 2025	828	925.0	17.8	16.1	327.3	25.2
San Juan	Jan 2025	862	921.4	17.5	16.4	328.3	25.6

Table S21. Level-by-level sounding data used for κ_{sonde} computation. T_d = dewpoint temperature derived from IGRA2 dewpoint depression field. N = Smith-Weintraub refractivity computed at each level. Wet% = wet term as percentage of total N . Singapore Jan 2025 data from prior session (SNM00048698, 7 levels, κ_{sonde} = 0.27611, wet% = 30%). IGRA2 source: NOAA NCEI (2023).

S7.3 OLS Regression Results and κ_{sonde} Derivation

Table S22 presents the OLS regression statistics for each of the six soundings, including the slope (dN/dz), R^2 of the linear fit, and the derived κ_{sonde} . High R^2 values confirm that the refractivity gradient is well-represented by linear regression over the 0–500 m analysis window.

Station	Date	N levels	dN/dz (N/km)	R^2	κ_{sonde}	Wet% surface
Barbados	Jul 2025	7	−52.50	0.971	0.33448	30%
Barbados	Jan 2025	8	−40.24	0.987	0.25635	28%
Pohnpei	Jul 2025	6	−44.62	0.994	0.28441	33%
Pohnpei	Jan 2025	8	−48.18	0.963	0.30703	34%
San Juan	Jul 2025	8	−48.72	0.982	0.31017	28%

San Juan	Jan 2025	6	−33.65	0.956	0.21441	26%
Singapore	Jan 2025	7	−43.35	0.978	0.27611	30%

Table S22. OLS regression statistics for κ sonde derivation. dN/dz = slope of N vs altitude regression. R^2 = coefficient of determination. All $R^2 > 0.95$ confirm linear refractivity profile assumption is valid over the 0–500 m window. $\kappa_{\text{sonde}} = -(R \times dN/dz \times 10^{-6})$, $R = 6,371,000$ m.

S7.4 Complete ERA5 vs Sonde Comparison

Table S23 presents the complete side-by-side comparison of ERA5-derived κ (from KAPPA API) and sonde-derived κ (from Table S22) for all six comparisons, including the ERA5 API response details.

Station	Date	κ_{ERA5}	dN/dz ERA5 (N/km)	ERA5 RH%	κ_{sonde}	$\Delta\kappa$	Gap%	Rating
Barbados	Jul 2025	0.27069	−42.49	86%	0.33448	−0.064	21.1%	POOR
Barbados	Jan 2025	0.26563	−41.69	86%	0.25635	+0.009	3.6%	EXCL.
Pohnpei	Jul 2025	0.27381	−42.98	87%	0.28441	−0.011	3.8%	EXCL.
Pohnpei	Jan 2025	0.26689	−41.89	81%	0.30703	−0.040	14.0%	FAIR
San Juan	Jul 2025	0.26599	−41.75	84%	0.31017	−0.044	15.3%	FAIR
San Juan	Jan 2025	0.26087	−40.95	83%	0.21441	+0.046	19.5%	FAIR
Singapore	Jan 2025	0.29149	−45.73	88%	0.27611	+0.015	5.4%	GOOD
MEAN	—	—	—	—	—	—	10.5%	—

Table S23. Complete ERA5 vs IGRA2 radiosonde κ comparison — all seven high-density tropical comparisons including Singapore Jan 2025 from prior session. ERA5 κ from KAPPA API /api/compute/elite endpoint at exact station coordinates. ERA5 RH% = relative humidity at surface layer from ERA5 response. Gap% = $|\kappa_{\text{ERA5}} - \kappa_{\text{sonde}}| / \text{mean}(\kappa_{\text{ERA5}}, \kappa_{\text{sonde}}) \times 100\%$. Singapore included here for completeness; Table 8 in main paper uses 6 comparisons at 3 stations $\times$ 2 seasons for methodological consistency. Rating: EXCELLENT <5%, GOOD 5–10%, FAIR 10–20%, POOR >20%.

S7.5 Barbados July 2025 — Tropical Wave Meteorological Documentation

The Barbados July 2025 comparison yields the largest gap (21.1%, POOR). Section 4.4.3 of the main paper attributes this to a documented tropical wave passage on 15 July 2025. The following meteorological records confirm the event:

Source	Time	Statement	Significance
CBC Barbados / Barbados Met Services	Morning (00Z)	'The tail-end of a tropical wave is affecting the island'	Active wave present at sonde launch time
CBC Barbados / Barbados Met Services	Evening (12Z)	'Another tropical wave will begin to affect the island'	Second wave arriving — double wave event
ERA5 analysis	00Z–06Z	$\kappa_{\text{ERA5}} = 0.27069$ (6-hourly average)	Smooths over wave-

			induced moisture surge
IGRA2 sonde	00Z (launch)	$\kappa_{\text{sonde}} = 0.33448$ (instantaneous)	Captures peak boundary layer moistening

Table S24. Meteorological documentation for Barbados 15 July 2025 tropical wave event. CBC Barbados source: <https://www.cbc.bb/news/weather-news/weather-forecast-for-tuesday-july-15-2025/> (CBC Barbados, 2025). The 21.1% gap is attributed to ERA5 6-hourly temporal averaging across the wave passage, not systematic wet-term deficiency.